

AQUASORT V1.1: IOT DUAL TANK WATER CLASSIFICATION

Automated Real Time Batch Testing & SLS 614:2013 / WHO Potable Water Compliance Engine

Microcontroller: ESP32 + 16-Bit ADS1115

UI: Interactive Web & Mobile Dashboard

Target Standard: Sri Lanka SLS 614

1. THE PROBLEM

Household water supplies face unannounced contamination and severe sediment spikes. Pumping contaminated water directly into bulk overhead tanks corrupts the entire domestic reserve. Furthermore, inline probes suffer from high drift, hydrodynamic turbulence, micro-air bubble noise and crosstalk.

2. PROPOSED SOLUTION

An IoT-enabled gatekeeping system featuring a 2-tier mechanical sampling chamber (1-2L vent + 10L acrylic sampling tank (T Sample). A 5 sensor array evaluates batch quality under static hold conditions, actuating zero differential solenoid valves to route Compliant water to **Tank 1 (Drinking)** or non-compliant water to **Tank 2 (Utility)**.

3. SYSTEM ARCHITECTURE & FLUID WORKFLOW

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[Inlet Supply] → [Stabilization Chamber (1-2L)] → [Sampling Tank T sample (10L)] → [5 Sensor Array + ADS1115 16-Bit ADC]
└────────→ [ESP32 4-State Machine Engine] → [Optocoupled Relay Driver] → [V1: Tank 1 (Drinking)] / [V2: Tank 2 (Utility)]
└────────→ (MQTT over TLS) → [IoT Interactive Web Dashboard & Cumulative Analytics Engine]
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4. CURRENT PROGRESS STATUS

- Mechanical Design:** 2 tier acrylic sampling chamber specifications and valve placement finalized.
- Firmware:** ESP32 4 state machine control logic (FILL, STABILIZE, SAMPLE, DRAIN) written and simulated.
- IoT Web UI:** Web app dashboard frontend in active development (MQTT telemetry, SLS 614 preset selector, cumulative tank running average calculator).

5. TECHNOLOGY STACK

Sensors: E-201-C pH, Titanium TDS, TSD-10 Turbidity, Platinum ORP, DS18B20 Temp, XKC-Y25 Level, JSN-SR04T Ultrasonic

Hardware: ESP32 Devkit V1, ADS1115 16-Bit ADC, 12V Solenoid Valves

Software: C++ Firmware, MQTT over TLS, Web App Dashboard

6. TESTING & CALIBRATION RULES

Validated against real world groundwater contamination events

SLS 614 Threshold Matrix:

- pH:** 6.5 – 8.5 | **TDS:** < 500 mg/L
- Turbidity:** < 2.0 NTU | **ORP:** > +200 mV
- ATC:** Real-time DS18B20 compensation via ADS1115 ADC.

7. RISKS & NEXT ACTION PLAN

Known Risks & Mitigation:

- Probe Drift:** Solved via spatial probe isolation & periodic web calibration.
- Ultrasonic Blind Zone:** 25 cm PVC riser necks added to T1/T2 tanks.
- Oct 6–8:** Physical acrylic tank fabrication & ADS1115 ADC calibration.
- Oct 9–11:** End to end cloud MQTT integration & live physical batch sorting validation.